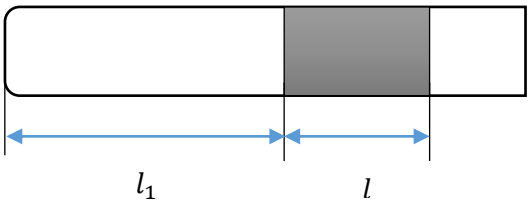
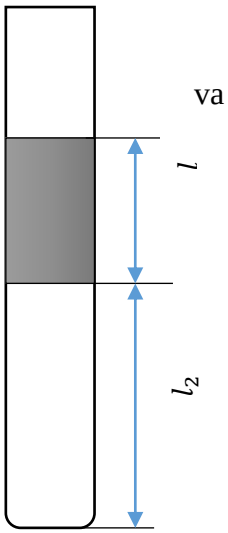
1- Masala

Bir uchi kavsharlangan shisha turubka ichidagi havo ustuni uzunligi l_1=307mm bo'lib, havo uzunligi l=216 mm bo'lgan simob ustuni bilan qamalgan.Naycha a) ochiq uchi yuqoriga b) ochiq uchi pastga qilib vertikal joylashtirilsa havo ustuni uzunligi qancha bo'ladi? Simob naydan to'kilmaydi. Atmosfera bosimi 747 mm simob ustuniga, simobning zichligi 13,6 g/ sm^3ga teng



a shartga ko'ra ochiq uchi yuqoriga qaraganda idish ichidagi bosim atmosfera bosimi va simob bosimlarining yig'indisidan iborat bo'ladi.



dastlabgi holatda havo bosimi $P_1 = P_0$ hajmi esa $V_1 = s * l_1$ bo'ladi.

$P_2 = P_0 + \rho g l$ hajmi esa $V_2 = s * l_2$ bo'ladi

Temperatura o'zgarmas ekanligidan

$$P_1V_1 = P_2V_2$$ kelib chiqadi

$$P_0 * s * l_1 = (P_0 + \rho g l) * s * l_2$$

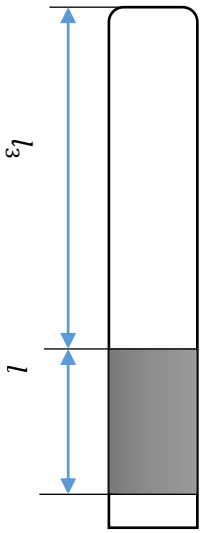
Hisoblashlar oson bo'lishi uchun bosimlarni mm sim ustlarda ifodalaymiz

$$747 * s * 307 = (747 + 216) * s * l_2$$ bu tenglamadan l_2 ni topsak

$$l_2 = \frac{(747*s*307)}{(747+216)*s}$$ =238,14 mm

b shartga ko'ra. Yopiq uchi bilan pastga qaratilganda esa umumiy bosim atmosfera bosimiga teng bo'lsagina suyuqlik butunlay to'kilib ketmaydi.

dastlabgi holatda havo bosimi $P_1 = P_0$ hajmi esa $V_1 = s * l_1$ bo'ladi. b shartda esa



$P_3 = P_0 - \rho g l$ hajmi esa $V_3 = s * l_3$ bo'ladi

Temperatura o'zgarmas ekanligidan

$$P_1V_1 = P_3V_3$$ kelib chiqadi

$$P_0 * s * l_1 = (P_0 - \rho g l) * s * l_3$$

Hisoblashlar oson bo'lishi uchun bosimlarni mm sim ustlarda ifodalaymiz.

$$747 * s * 307 = (747 - 216) * s * l_3$$ bu tenglamadan l_3 ni topsak

$$l_3 = \frac{(747 * s * 307)}{(747 - 216) * s} = 431{,}88mm$$

Kelib chiqadi.

$$javob: l_2 = 238{,}14 \text{ mm}, l_3 = 431{,}88mm$$

2- Masala

Radiusi R_1 bo'lgan sovun pufagi va radiusi R_2 bo'lgan sovun pufagi birlashib radiusi R bo'lgan pufakni hosil qildi. Atmosfera bosimi P_0 bo'lsa pufakning sirt taranglik koeffitsiyentini toping.

Yechim:

Dastlabgi holatda ikkala pufak ham atmosfera bosimida turibdi. Bu pufaklarda ichidagi havoning ichki energiyasi va pufakning sirt energiyalari bor. Havoni bir atomli gaz deb qarasak.

$$W_1 = 2\delta S_1 + \frac{3}{2}P_0V_1 = 2\delta 4\pi {R_1}^2 + \frac{3}{2}P_0\frac{4}{3}\pi {R_1}^3 = 2\pi {R_1}^2(4\delta + P_0R_1)$$

$$W_2 = 2\delta S_2 + \frac{3}{2}P_0V_2 = 2\delta 4\pi {R_2}^2 + \frac{3}{2}P_0\frac{4}{3}\pi {R_2}^3 = 2\pi {R_2}^2(4\delta + P_0R_2)$$

$$W_N = 2\delta S + \frac{3}{2}P_0V = 2\delta 4\pi R^2 + \frac{3}{2}P_0\frac{4}{3}\pi R^3 = 2\pi R^2(4\delta + P_0R)$$

Energiyaning saqlanish qonuniga ko'ra

$$W_1 + W_2 = W_N$$

$$2\pi {R_1}^2(4\delta + P_0R_1) + 2\pi {R_2}^2(4\delta + P_0R_2) = 2\pi R^2(4\delta + P_0R)$$

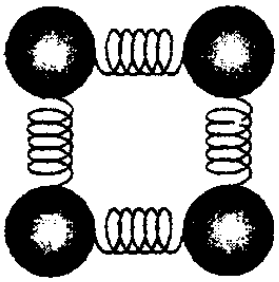
$${R_1}^2(4\delta + P_0R_1) + {R_2}^2(4\delta + P_0R_2) = R^2(4\delta + P_0R)$$

$$4\delta\left({R_1}^2 + {R_2}^2 - R^2\right) = P_0(R^3 - {R_1}^3 - {R_2}^3)$$

$$\delta = \frac{P_0\left(R^3 - {R_1}^3 - {R_2}^3\right)}{4({R_1}^2 + {R_2}^2 - R^2)}$$

$$javob: = \frac{P_0\left(R^3 - {R_1}^3 - {R_2}^3\right)}{4({R_1}^2 + {R_2}^2 - R^2)}$$

3- Masala



Agar har bir prujinaning bikrligi K ga teng bo'lsa va har bir jismning massasi m ga teng bo'lsa sistemaning tebranish davrini toping.

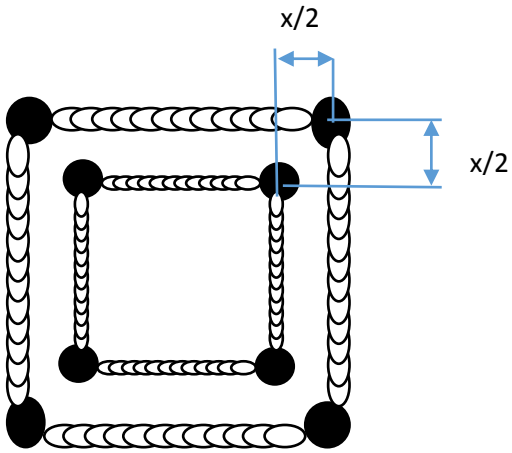
Energiyaning saqlanish qonuniga ko'ra

$$\frac{4mv^2}{2} = \frac{4kx^2}{2}$$

$$A = \frac{x}{2}\sqrt{2}$$

$$x = v\sqrt{\frac{m}{k}}$$

$$A = \frac{x}{2}\sqrt{2} = v\sqrt{\frac{m}{2k}}$$



Agar Siljishni sinus qonuniyati bo'yicha o'zgaradi deb hisoblab tenglamalarni yozamiz.

$$X = A\sin\omega t \qquad (1)$$

Siljishdan vaqt bo'yicha olingan birinchi tartibli hosila tezlikning o'zgarish tenglamasini ifodalaydi.

$$\dot{x} = v = A\omega\cos\omega t \qquad (2)$$

Bu tenglamadan maksimal tezlik

$$v_{max} = A\omega \quad (3)$$

ekanligini bilib olishimiz mumkin

$v_{max} = A\omega$ bu tenglamadan

$$\omega = \frac{v_{max}}{A} = \frac{v}{v\sqrt{\frac{m}{2k}}} = \sqrt{\frac{2k}{m}}$$

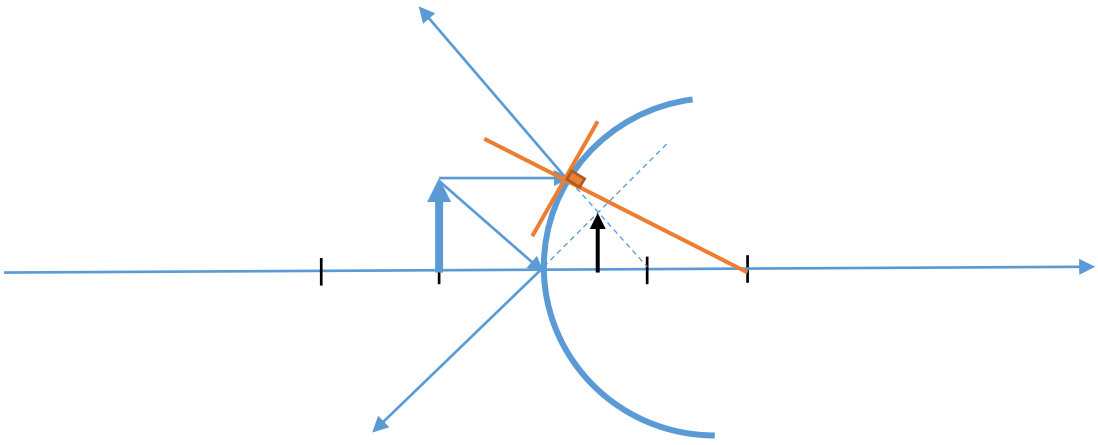
$$T = \frac{2\pi}{\omega} = 2\pi\sqrt{\frac{m}{2k}} = \pi\sqrt{\frac{2m}{k}}$$

$$javob: T = \pi\sqrt{\frac{2m}{k}}$$

4- Masala

Qavariq ko'zguning egrilik radiusi 80 sm ga teng. Predmetdan ko'zgugacha bo'lgan masofa 40 sm ga teng. Tasvirning chiziqli kattalashtirishini aniqlang. Va tasvirdan buyumgacha masofa qanday bo'ladi.

R=2F ekanligidan F=40 sm ekanligi kelib chiqadi. Demak d=F ekan



Qabariq ko’zguda doimo kichiklashgan mavhum va to’g’ri tasvir hosil bo’ladi va qabariq ko’zgu sochuvchi linza formulasiga tushadi.

$$-\frac{1}{F}=\frac{1}{d}-\frac{1}{f}$$

$$-\frac{1}{0,4}=\frac{1}{0,4}-\frac{1}{f}$$

$$\frac{1}{f}=\frac{1}{0,4}+\frac{1}{0,4}=5$$

$$f=20\ sm\ bo'ladi$$

$$k=\frac{f}{d}=\frac{0,2}{0,4}=0,5$$

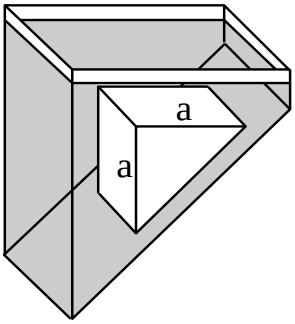
Buyumdan tasvirgacha masofa esa $l=f+d=0,6\ m=60\ sm$

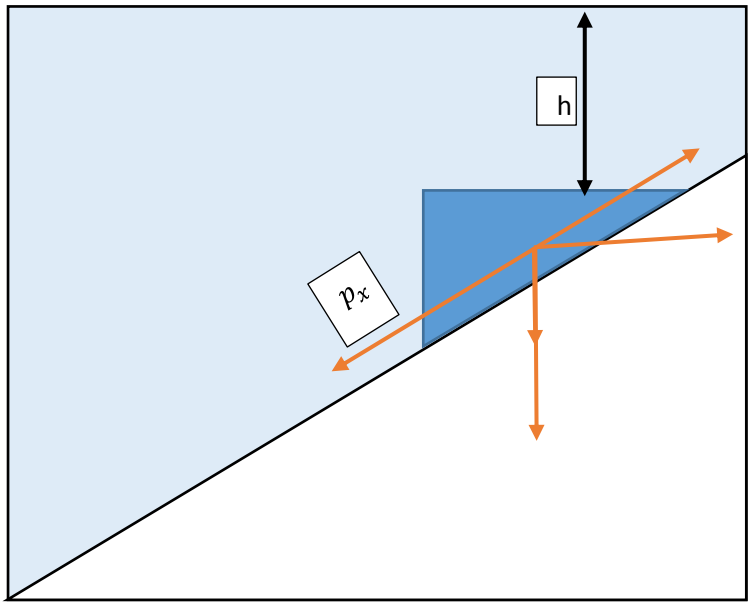
$$javob: k=0,5 \quad va \quad l=60\ sm$$

5- Masala

Qirrasining uzunligi a=40 sm bo`lgan kubning yarmi (prizma) rasmda ko`rsatilgani kabi suvli idishning qiya sirtiga qo`yilgan va uning tagiga suv kirmaydi. Agar prizmaning yuqori sirti suv sathidan 20 sm pastda va prizma bilan idishning qiya sirti orasidagi ishqalanish koeffitsiyenti 0,2 ga teng bo`lsa, prizma muvozanatda bo`lishi uchun u tayyorlangan moddaning zichligi qanday bo`lishi kerak (kg/m^3)?

(P_o=10^5 Pa)





$jismga\ ta'sirqiluvchi\ kuchlarning\ proyeksiyalarini\ ko'ramiz$

Jismning sirtga ko’rsatuvchi bosim kuchini quyidagicha topamiz.

$$N = \left(mg + (p_0 + \rho gh) * s + \frac{\rho ga}{2}\right) * cos\alpha$$

Jismga ta’sir etuvchi barcha kuchlarning proyeksiyasini ko’rsak

$$\mu\left(mg + (p_0 + \rho gh) * s + \frac{\rho ga}{2}\right) * cos\alpha = mg * sin\alpha + (p_0 + \rho gh)s * sin\alpha + \frac{\rho gas}{2}$$

Hisoblashlarni amalga oshirsak quyidagicha bo’ladi:

$$0{,}8\left(\rho_j * \frac{4^3}{10^3} * 10 + (10^5 + 10^4 * 0{,}2) * 0{,}16\right) = 1{,}2(10{,}4 * 0{,}16 * 100)$$

$$0{,}02\rho_j = 540$$

$$\rho_j = 27000\ \frac{kg}{m^3}$$

@olimpdep



Fan olimpiadalari bo‘yicha
iqtidorli o‘quvchilar bilan ishlash
DEPARTAMENTI